The Bayesian recipe

Outline

Topics

- Introduce the Bayesian Recipe (synonym for the Bayes estimator)
- Illustrate it using this week's running example

Rationale

The Bayesian Recipe/Bayes estimator is the guide for all "full Bayesian" statistical analyses. This week we apply it to an example that builds on last week's review.

Preview of key definitions

Bayesian Analysis: statistical discipline centered around the use of **Bayes estimators**.

Bayes estimators: for data Y, unobserved X, loss L, and possible actions A, the Bayes estimator is defined as:

$$\arg\min\{\mathbb{E}[L(a, X)|Y] : a \in A\}$$

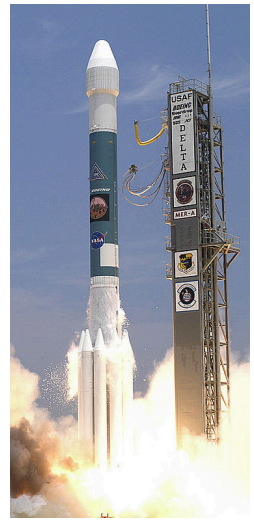
Note: you are not expected to understand this equation at this point!

This week's running example

- You are consulting for a satellite operator
- They are about to send a $100M satellite on a Delta 7925H rocket

- **Data**: as of Jan 2025, Delta 7925H rockets have been launched 3 times, with 0 failed launches
  - Note: Delta 7925H is not reusable, so each rocket is "copy- built" from the same blueprint
- Should you recommend buying a $2M insurance policy?

Convention: use 1 for a success, 0 for a failure.



The Bayesian recipe

The goal this week is to undersand the 3 steps in the Bayesian recipe:

1. Construct a **probability model** including
   - random variables for what we will measure/observe
   - random variables for the unknown quantities $\theta = P(\text{success})$
     - those we are interested in ("parameters", "predictions")
     - others that just help us formulate the problem ("nuisance", "random effects"). $\Rightarrow$ prior
2. Compute the **posterior distribution** (condition on the data) $P(\theta|\text{data})$ $\propto$ likelihood $\times$ prior $P(\text{data}|\theta)$
3. Use the posterior distribution to (**decision theory**):
   - make prediction (point estimate) success rate
   - estimate uncertainty (credible intervals)
   - make a decision

Plan

- Understanding Step 1 ("construct a probability model"):
  - we reviewed probability models last week,
  - in fact, the model we use for this week's problem is the same as last week's!
  - We will just need to add some Bayesian terminology: **prior, likelihood**.
- Understanding Step 2 ("condition on the data"):
  - we reviewed conditional probability last week,
  - in fact, the conditional probability calculation for this week's problem is the same as last week's!
  - We just need to add some Bayesian terminology: **posterior distributions**.
- Understanding Step 3: this is where most of the new material will be for this week.

Bayesian models

Outline

Topics

- Bayesian interpretation of probability models: from aleatoric to epistemic.
- Terminology: prior, likelihood, joint.

Rationale

This week, we will give a new interpretation to the "bag of coin" example (where the uncertainty is "aleatoric"), turning it into a Bayesian model (where the uncertainty is "epistemic"). Mathematically this is the same model but with a different interpretation. This shift of interpretation is the basis of all Bayesian models.

What is a Bayesian model?

X随机变量.

A **Bayesian model** is a probability model equipped with:

- random variable(s) representing the data (we use Y in this course)
- random variable(s) for the unknown quantities (we use X in this course).

Note: concretely, a Bayesian model is a joint PMF over X and Y, typically written using the "$\sim$" notation introduced last week.

From coins to rockets

For this week's rocket example, we will use the exact same probability model as the bag of coin example from last week.

Use the following correspondence to link last week's example with this week's:

- $(Y_i = 1)$:
  - $\leftrightarrow$ "i-th flip is a heads"
  - $\leftrightarrow$ "i-th launch is a success"
- $(X = k)$:
  - $\leftrightarrow$ "the coin drawn from bag has probability $p = k/K$ of heads"
  - $\leftrightarrow$ "this type of rocket has probability $p = k/K$ of success."

Terminology

The task of constructing a Bayesian model is often broken down into constructing:

$P(X,Y) = P(Y|X)P(X)$.

- the **prior**: the PMF of the unknown X,[1] $P(X)$ #X的分布
- the **likelihood**: the conditional PMF of observed Y given X. $P(Y|X)$ #Y的分布 Bern(·).

Epistemic vs aleatoric probability

- In the bag of coin example, we can replicate the whole "experiment" several times (both prior and likelihood).
- In the rocket example, we can only replicate launches (likelihood), not the prior sampling part.

因为无法再模拟出1个完全相同的rocket.

Terminology:

认知的不确定性.

- **Epistemic probability**: the part we cannot replicate. 知道真相后就会消失. ⇒ prior/posterior.
- **Aleatoric probability**: the part that can be replicated. 知道一切也无法消掉. ⇒ likelihood

随机性本身

Uses:

- Bayesian statistics uses both types of probability,
- whereas other fields of statistics, e.g. MLE, typically uses only aleatoric.

Footnotes

1. if the unknown quantity is continuous, the prior will be expressed using a density. A term that captures both the continuous and discrete case is "distribution" i.e. "prior distribution". ↩

Posterior distributions

Outline

Topics

- Notion of posterior distribution.
- Examples
- How to use a posterior?

Rationale

The posterior distribution appears in the second step of the Bayesian Recipe and is therefore encountered (at least implicitly) in all full Bayesian problems.

Definition

The conditional PMF of the unknowns X given the observation Y is the called the **posterior PMF**.[1]

Notation

这本质就是 Bayes 定理：

$$P(X = z \mid Y = y) = \frac{P(Y = y \mid X = z)\, P(X = z)}{P(Y = y)}$$

unknown.

We use $\pi(x) = \mathbb{P}(X = x | Y = y)$ for the posterior PMF.

known.

Recall:

$$\pi(x) = \frac{\gamma(x)}{Z},$$

$= P(Y=y|X=z)P(X=z)$

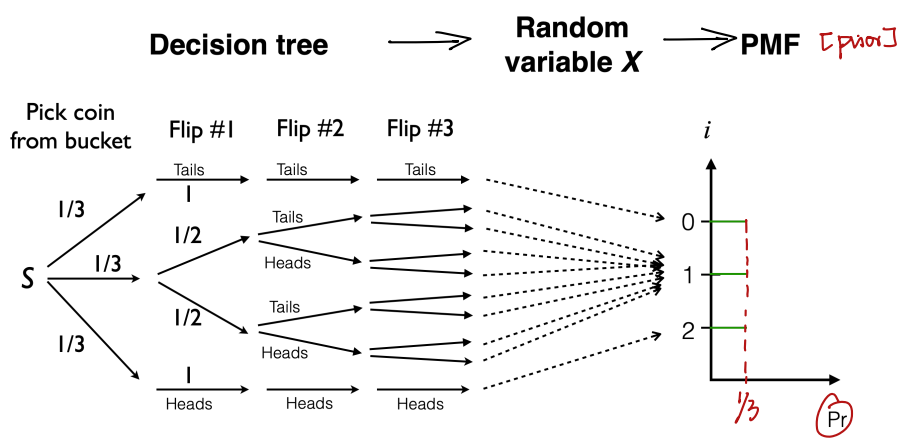
where $\gamma(x) = \mathbb{P}(X = x, Y = y)$ is the un-normalized posterior and $Z = \mathbb{P}(Y = y)$ is the normalization constant.

Examples

将和observed不相容的path抹掉再重新归一化.

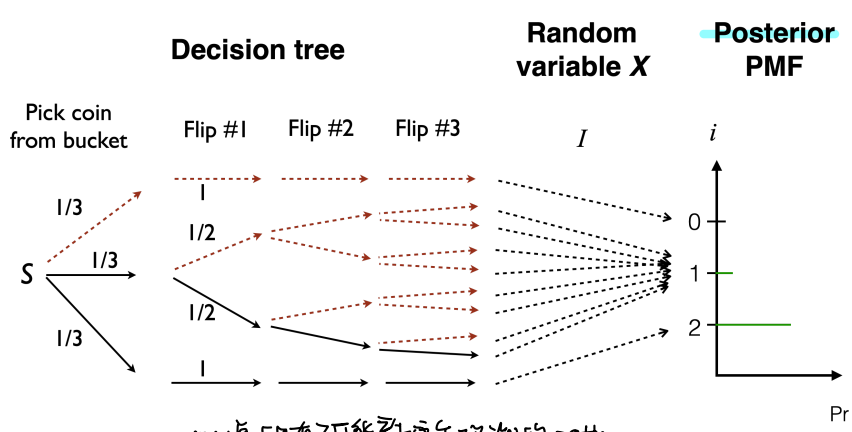
Example 1: coins in a bag

Visualization of the prior PMF and how it arises from the decision tree and random variable X (showing 3 flips instead of 4):



- Recall that the probability of each path is the product of the edge labels.

Visualization of the **posterior PMF** and how it arises from the decision tree and random variable X:



删掉所有不可能产生这个观测的path.

- Recall we zero out the contribution of the paths not compatible with the observation (Heads, Heads, Heads).
- This gives a list of numbers that do not sum to one, so we renormalize them. 归一化

Example 2: rocket insurance

You will construct prior and posterior PMFs in question 2 of this week's exercises.

What to do with a posterior?

- Show a visualiation (posterior PMF).
- Compute a **summary** of the PMF:
  - **point estimate:** single “best guess”, or
  - **a credible region:** a set of “guesses”.
- More generally: decision theory / Step 3 of the Bayesian Recipe. 量化不确定性并支持决策.

Footnotes

1. if the unknown quantity is continuous, the posterior will be expressed using a density. A term that captures both the continuous and discrete case is “distribution” i.e. “posterior distribution”. ↩

Point estimates

Outline

Topics

- Common point estimates:
  - Posterior mean.
  - Posterior mode.

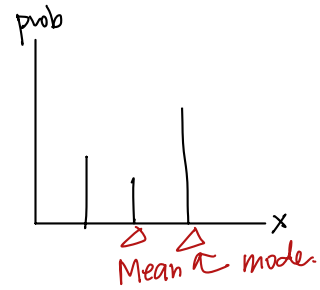
Rationale

It is often necessary to summarize the posterior distribution with a single "best guess", even though as we will see this hides important information namely our uncertainty about that guess.

Definitions

- Let $\pi(x) = \mathbb{P}(X = x | Y = y)$ denote a posterior PMF.
- **Point estimate:** Instead of plotting the full information in π, we can report a "location" summary such as the mean of the posterior π.

① Posterior mean



Recall the mean is computed from a PMF via

$$E[X|Y=y] = \sum_x x\,\pi(x),$$

where the sum is over $\{x : \pi(x) > 0\}$.

Notation: the posterior mean is denoted $\mathbb{E}[X|Y = y] = \sum x\,\pi(x)$.

② Posterior mode

The **mode** is the location of the "tallest stick" in the PMF.

Notation: $\arg\max \pi(x)$, i.e. the point that achieves the maximum of π. ex) posterior pmf 中 prob. max 되는 x.

In the Bayesian context, the mode of a posterior PMF is also known as the Maximum A Posteriori (MAP) estimator.

Examples

Example 1

In the following, we will compute point estimates based on $\pi(x) = \mathbb{P}(X = x | Y = (1,1))$ in the bag of coin running example.

$Y=1$

- Imagine a bag with 3 coins each with a different probability parameter p
- Coin $i \in \{0, 1, 2\}$ has bias $i/2$—in other words:
  - First coin: bias is $0/2 = 0$ (i.e. both sides are "heads", $p = 0$)
  - Second coin: bias is $1/2 = 0.5$ (i.e. standard coin, $p = 1/2$)
  - Third coin: bias is $2/2 = 1$ (i.e. both sides are "tails", $p = 1$)

- Consider the following two steps sampling process
  - Step 1: pick one of the three coins, but do not look at it!
  - Step 2: flip the coin 4 times
- Mathematically, this probability model can be written as follows:

$$\begin{aligned} X &\sim \text{Unif}\{0, 1, 2\} \\ Y_i | X &\sim \text{Bern}(X/2) \end{aligned} \tag{1}$$

1. un-normalized posterior

$\gamma(x) = P(Y=(1,1) | X=x) P(X=x)$

① $x=0$: $P(Y|X=0) = 0$

② $x=1$: $P(Y|X=1) P(X=1) = (1/2)^2 \cdot 1/3 = 1/12$

③ $x=2$: $P(Y|X=2) P(X=2) = 1/3$.

2. normalized post.

$Z = 0 + 1/12 + 1/3 = 5/12$

$\Rightarrow \pi(0) = 0,\ \pi(1) = \frac{1}{12} \cdot \frac{12}{5} = 1/5$

$\pi(2) = 4/5. = \frac{1}{3} \times \frac{12}{5} = 4/5$

3. post. Mean

$E[X|Y] = 0 + 1 \cdot 1/5 + 2 \cdot \frac{4}{5} = 1.8$

4. post. Mode (MAP)

$\arg\max_x \pi(x) = 2.$ diff.

Question: compute the posterior **mean** of π.

$* \sum x \pi(x) \neq \sum x \gamma(x)$.

① $\gamma = (\gamma(0), \gamma(1), \gamma(2))$

② $\pi = \gamma / Z$

③ $E[X|Y=(1,1)] = \sum x \pi(x)$

Click for choices

Click for answer

Question: compute the posterior **mode** of π.

Click for choices

Click for answer

–>

Example 2

You will practice computing the posterior mean/mode in question 2 of the exercises.

Credible sets ⇒ 用集合来表达uncertainty.

Outline

Topics

- Nominal coverage.
- Common credible sets:
  - Quantiles.
  - Highest-density sets.

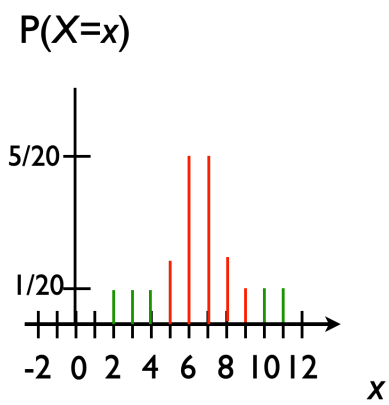
Rationale

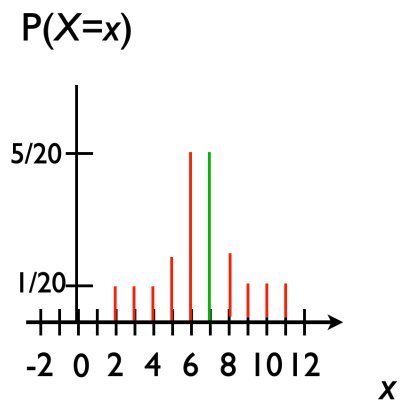
As we have seen at the beginning one motivation for Bayesian methods is that they allow us to quantify uncertainty in our predictions. Credible sets is one way to convey this uncertainty quantification.

Nominal coverage

The "nominal coverage" is the basic property that we use to construct credible sets:

- You are given a level, typically 95% or 90%, denoted $1-\alpha = 0.95$ or $1-\alpha = 0.9$
  - Intuition: we want to find a set T such that we are 95% sure that the true x is in T.
- Ideally, we would want to find a set T such that $\sum_{x \in T} p(x) = 0.9$
  - For discrete distributions, there may not be a solution, but let us ignore that issue for now (suppose each probability is very small and replace $=$ by $\approx$ up to an error bounded by the largest individual probability)
  - For continuous distributions that will not be an issue
- Another problem: there are many solutions! Some of them might appear weird, e.g. include points of probability zero!
  - Example: both sets shown in red contain 75% of the mass.





Method for building credible interval

① Quantile-based credible interval

The simplest way of building credible intervals:

- suppose you want a 90% credible set
- we will remove $10\%/2 = 5\%$ on each side
- remove sticks from the very left moving to right
  - each time you remove one, add its length to a counter…
  - until that counter hits 5%.
- then do the same from very right moving to the left
- the sticks left form T.

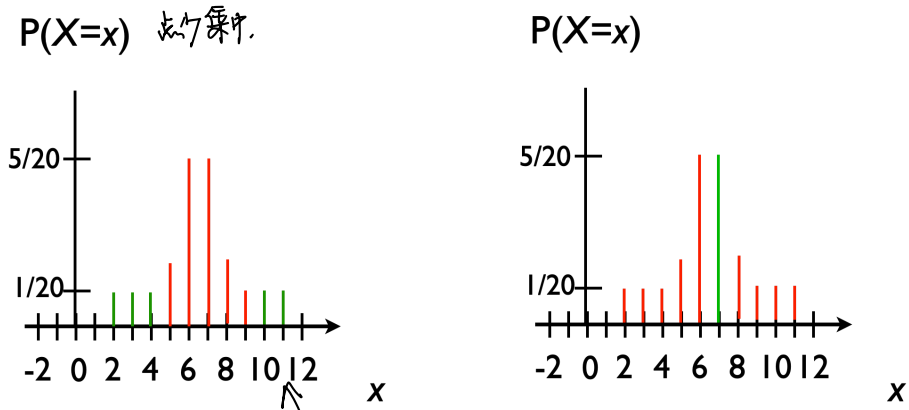
② Highest Density Set

① rank $\pi(x)$ 从小到大.
② 依次加入直到 $\sum p(x) \geq 1-\alpha$.

- **Highest Density Set**: pick a solution with the least number of points $|T|$
- To handle discrete models: $\operatorname{argmin}\{|T| : \sum_{x \in T} p(x) \geq 0.9\}$.
  - Still not unique in some corner cases, but good enough

Example:

- the left one below is a highest density set
- note the one on the right has more points in it, $5 < 9$.



点分布.

Challenge: implement a function taking as input a posterior PMF (as a vector), a level, and returns an HDI. Details here.

Credible interval

continuous的情况.

When we move to continuous random variables, T will often be an interval, i.e. $T = [L, R]$ for some left and right end points L and R. In such case, T is called a credible interval.

Highest Density Interval

Similarly, a Highest Density Interval (HDI) is a shortest interval (i.e. minimizing $R - L$) containing a prescribed probability mass.

argmin $L(a)$: 令$L(a)$最小的a值.
min $L(a)$: $L(a)$的最小值

Decision theory Min expected loss.

Outline

Topics

- Decision theory: making decision in the face of uncertainty.
  - First, without observed data.
  - Second, incorporating observed data.
- Review of property of expectation useful for decision theory calculations.

Rationale

We talked a lot about the "Bayesian recipe." Here we formalize the last step of that recipe: using a posterior and a loss function to make a decision.

Later, we will see that the point estimates and credible sets covered this week are actually special cases of the decision theoretic framework (i.e. they emerge when specific loss functions are selected).

Decision theory without data

Your friend proposes the following game (a good model for lottery!):

- She flips a standard coin:
  - if it's a "heads": you give her \$10. ⇒ 正损失 +10
  - if it's a "tails": she gives you \$9. ⇒ 负损失 −9
- Your decision: to play or not to play?

Mathematical formulation

- Set of **actions**: $A = \{0, 1\}$ ($1 =$ play, $0 =$ do not play).
- Probability model: $X \sim \text{Bern}(1/2)$.
- Loss function:
  - $L(a, x)$: how much do I lose if I pick action $a \in A$ and encounter realization $X = x$?
- Mathematical solution: minimize over $a \in A$ the **expected loss**

$$L(a) = \mathbb{E}[\underbrace{L(a, X)}_{g(x)}].$$

a: not random ⇒ A number.

- Recall, by LOTUS:

$$\mathbb{E}[L(a, X)] = \sum_x L(a, x) p_X(x).$$

a \ x	0	1
0	\$0	\$0
1	−\$9	\$10

$L(0) = 0$

$L(1) = \frac{1}{2} \cdot 10 + \frac{1}{2} \cdot (-9) = 1/2.$

Example

Compute $\arg\min L(a)$ and $\min L(a)$.

$\cdot(\$10)\}$

(x)

It is useful and important to answer this easy question in excruciating mathematical details. This will prepare us for more complex decision scenarios.

Let us start by writing the loss function mathematically using **indicator function** defined as:

$$\mathbb{1}[\text{some condition}] = \begin{cases} 1 & \text{if the condition is true,} \\ 0 & \text{if the condition is false.} \end{cases}$$

This gives us:

not play play

$$L(a, x) = \mathbb{1}[a = 0] \cdot (\$0) + \mathbb{1}[a = 1, x = 0] \cdot (-\$9) + \mathbb{1}[a = 1, x = 1] \cdot (\$10)$$

Let us simplify the expected loss function, i.e.

$$\mathbb{E}[L(a, X)] = \mathbb{E}[\mathbb{1}[a = 0] \cdot (\$0) + \mathbb{1}[a = 1, X = 0] \cdot (-\$9) + \mathbb{1}[a = 1, X = 1] \cdot (\$10)].$$

o do that simplification, first use **linearity of expectation** which you will recall from probability theory tells us that for any random variables X_1 and X_2, we have $\mathbb{E}[X_1 + cX_2] = \mathbb{E}[X_1] + c\mathbb{E}[X_2]$ for any constant c.

In our context, using linearity with $c = 1$ allows us to simplify the above to:

$$\underbrace{\mathbb{E}[\mathbb{1}[a = 0] \cdot (\$0)]}_{0} + \mathbb{E}[\mathbb{1}[a = 1, X = 0] \cdot (-\$9)] + \mathbb{E}[\mathbb{1}[a = 1, X = 1] \cdot (\$10)].$$

Trick 1: To deal with each remaining term, use that: $\mathbb{1}[a = 1, X = 1] = \mathbb{1}[a = 1]\mathbb{1}[X = 1]$ (check!), and then use linearity with $c = \$10 \cdot \mathbb{1}[a = 1]$ to get:

$$\mathbb{E}[\mathbb{1}[a = 1, X = 1] \cdot (\$10)] = \$10 \cdot \mathbb{1}[a = 1]\mathbb{E}[\mathbb{1}[X = 1]].$$

LOTUS: Finally, we have to compute $\mathbb{E}[\mathbb{1}[X = 1]]$. From our review of the Law of the Unconscious Statistician, this is done as follows:

$$\mathbb{E}[\mathbb{1}[X = 1]] = \sum_x \mathbb{1}[x = 1] p_X(x) = 0 + 1 \cdot p_X(1) = 1/2.$$

Trick 2: you will often take expectation of indicator function, you will always find that $\mathbb{E}[\mathbb{1}[\text{an event}]] = \mathbb{P}(\text{an event})$.

Putting it all together:

$$L(a) = \mathbb{E}[L(a, X)] = -\$9 \cdot \mathbb{1}[a = 1](1/2) + \$10 \cdot \mathbb{1}[a = 1](1/2) = \$0.5 \cdot \mathbb{1}[a = 1].$$

$\begin{cases} L(0) = 0 \\ L(1) = 0.5 \end{cases} \Rightarrow L(0) < L(1)$

$\Rightarrow$ not to play

Decision theory with data

- We now consider the case where we have some data Y that give us information about X.
- The actions A and loss $L(a, x)$ are the same as before.

How to modify the last section to take into account the data Y?

- Do everything the same, except:
- use the conditional PMF $p_{X|Y}$ instead of the PMF p_X.

Definition: the Bayes estimator

Mathematically: use an action $a \in A$ minimizing the posterior expected loss, i.e.

$$\arg\min \mathbb{E}[L(a, X)|Y = y].$$

where:

$$\mathbb{E}[L(a, X)|Y = y] = \sum_x \boxed{L(a, x)}\boxed{p_{X|Y}(x|y)}.$$

Example

You will apply the Bayes estimator in question 3 of this week's exercises.

Prediction

Outline

Topics

- Prediction using decision trees
- Example

Rationale

Often we do not care so much about "parameters" but instead about predicting future observations.

Example: coins in a bag

Consider the setup from last week with 3 coins and 3 flips with $Y = (1, 1, 1)$ (in the following, let $\mathbf{1}$ denote a vector of 1's).

Question: given you have see 3 heads, what is the probability that the next one is also heads?

Mathematically: $\mathbb{P}(Y_4 = 1 | Y_{1:3} = \mathbf{1})$. This is known as "prediction".

General approach

Key message: In Bayesian statistics, prediction and parameter estimation are treated in the exact same way!

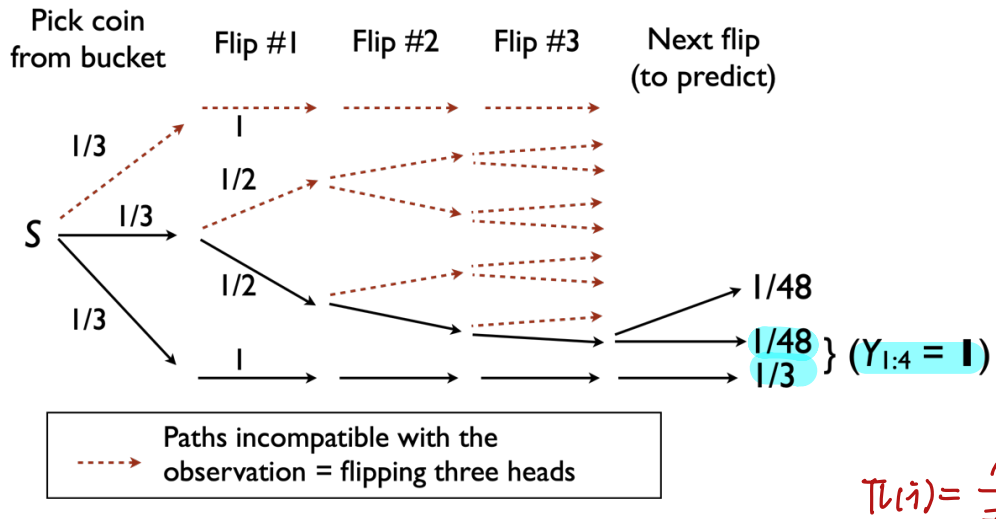
将Y4当作未知量,和coin type X一起处理.

Idea: Add Y_4 to the unobserved random variables, i.e. set $\tilde{X} = (X, Y_4)$.

Then, to compute $\mathbb{P}(Y_4 = 1 | Y_{1:3} = \mathbf{1})$ use same techniques as last week (decision tree, chain rule, axioms of probability).

Example, continued

Use the following picture to help you computing $\mathbb{P}(Y_4 = 1 | Y_{1:3} = \mathbf{1})$.



$$\pi(i) = \frac{r(i)}{Z\, r(i)}$$

$$r(i) = p(Y_{1:3}=1, Y_4=i)$$

$$(Y_{1:3}=1, Y_4=1) = (X=1, Y_{1:3}=1, Y_4=1)$$

$$\Rightarrow r(0) = p(Y_{1:3}=1, Y_4=0) = (1/3)\times(1/2)^4 = 1/48 \approx 0.02.$$

$$(Y_{1:4}=1) = (X=2, Y_{1:4}=1) \cup (X=1, Y_{1:4}=1)$$

$$\Rightarrow r(1) = p(Y_{1:3}=1, Y_4=1) = [1/3 \times (1)^4] + [1/3 \times (\tfrac{1}{2})^4]$$

$$= 1/3 + 1/48 = 17/48 \approx 0.35$$

$$\Rightarrow \underline{\text{predictive}}: \; P(Y_4=1 \mid \overset{\text{data}}{\boxed{Y_{1:3}=1}}) = \frac{P(Y_4=1, Y_{1:3}=1)}{P(Y_{1:3}=1)} = \frac{0.35}{0.02+0.35} \approx 0.94$$

$$\# \; \pi(i) := P(Y_4=i \mid Y_{1:3}=1)$$

$$\pi(i) \propto r(i)$$

$$(\pi(0), \pi(1)) = \frac{(r(0), r(1))}{r(0)+r(1)}$$

$$\Rightarrow P(Y_4=1 \mid Y_{1:3}=1) = \frac{r(1)}{r(0)+r(1)}$$

Intro to model criticism

模型批判.

Outline

Topics

- What is a model mis-specification?
- What are the consequences?
- What to do about it.

Rationale

Model mis-specification (the model being "too wrong") can lead to serious problems in all types of statistical models, but Bayesian models are often more seriously affected. As a result it is important to detect serious cases of mis-specification and to address them. This is just an introduction, we will go in more depth once we have introduced Bayesian regression models.

Definitions

结构性错误.

Model mis-specification: when the model is "too wrong" (in the context of the famous quote by George Box).

批判过程.

Model criticism: the task of trying to find mis-specification.

- It can be done by reasoning and discussion with experts.
- It can be done using data (**goodness-of-fit**)

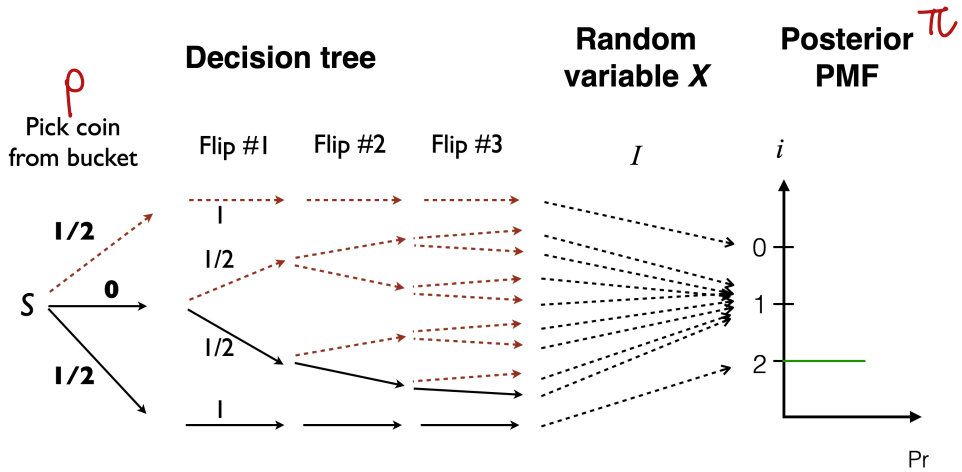
Example

Recall our notation: p prior PMF, π posterior PMF.

- Suppose we put zero prior mass to having a fair dice. What happens on the posterior?

$p(X=1)=0.$

$\Rightarrow$ 在Model中提前宣称某个情况不可能，data永远无法推翻你.



- If we put prior mass of zero to some realization x, say $\rho(x) = 0$ then the posterior probability on x will always be zero, $\pi(x) = 0$, *no matter how many observations we get*
  - This could be disastrous! (e.g. ignoring extreme scenarios can have extreme consequences)

不要将任何图谱上可能发生的事的prior prob.设为0.

- Principle to avoid this is known as **Cromwell's rule** (Oliver Cromwell, 1650): "I beseech you, in the bowels of Christ, think it possible that you may be mistaken." ⇒ 只要不是不可能，就不可以让其prior=0.

Discussion: can you identify other issues with the model from Exercise 2?

How to correct mis-specification

- Improve the model!
- Then iterate with more criticism and improve again if needed.
- Iterative model improvement is part of the current "best practice" in Bayesian analysis.